

Synthetic boundary layout sample

This is an artificial train only case, not a model result. The held-out-dose predictive criterion is undefined for all nine weight-space features; these comparisons cannot establish either support or lack of support. All nine features and all four folds are retained. Undefined folds are not pooled as available cases. This is prediction outside a learning-rate dose on the same task suite, not causal mediation, statistical significance or evidence of equivalence.

Geometry feature	RMSE reduction	Improved	Defined	Criterion	Residual ρ
log segment/weight norm	—	2/4	3/4	undefined	+0.697
segment stable-rank fraction	—	2/4	3/4	undefined	+0.697
segment effective-rank fraction	—	2/4	3/4	undefined	+0.697
segment row-norm CV	—	2/4	3/4	undefined	+0.697
segment top-1% row energy	—	2/4	3/4	undefined	+0.697
cumulative/weight norm	—	2/4	3/4	undefined	+0.697
cumulative stable-rank fraction	—	2/4	3/4	undefined	+0.697
segment subspace overlap to AdamW	—	2/4	3/4	undefined	+0.697
cumulative subspace overlap to AdamW	—	2/4	3/4	undefined	+0.697

Table 1: Every original weight-space predictor under four-fold leave-dose-index-out prediction. Decisions use exact MSE comparisons, not displayed RMSE. A missing correlation is undefined, not zero; baseline equivalence is distinct from an unidentified or numerically unresolved prediction.